

KOHLER

AquaSense OS

Real-Time Hydro-Intelligence for Commercial Facilities

Prompt Engineering Documentation

Submission Artifact: Prompts_Documentation.pdf

Version: 1.2

Deployment: Pune International Airport (PNQ)- NITB Departure Zone 3

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Phase 1- Problem Framing & Domain Grounding

1. *"Im building an AI-powered smart plumbing monitoring system for Kohler commercial fixtures deployed in a highfootfall airport terminal (pune international aiirport). The fixtures include touchless faucets, urinals, and toilet stalls. what are the most common failure modes for each fixture type in a commercial airport context, their typical detection lag without intelligence, and the water loss quantities associated with each? give me Kohler specific terminology where possible."*
2. *"1. streamlit db isnt much customizable especially font size; any other alternative?
2. for the prompts documentation which will be heavily evaluated, do i document every prompt? give an example"*
3. *"give an example of how this project would work from an end user's pov"*
4. *"what telemetry fields would a smart fixture realistically have? give me realistic data with units, typical normal ranges, and what anomalous values look like for each."*
5. *"what does a facility maintenance technician at an indian airport actually need to see on a work order to act immediately? give me the workflow from anomaly detection to physical repair."*
6. *"do we need any simulator in this project to make it more visual?"*
7. *"has this already been implemented by kohler?"*
8. *"which of these llms should i use for specific work in the project- claude, gemini, gpt? i have the free plans of all. also give the repo structure"*

9. *"before I implement this, what are the 3 most likely ways this will break in a live demo environment? Give work arounds for each beforehand"*
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Phase 2- UML Diagram

10. *"give me the uml code that i can paste into draw.io and get the uml class diagram"*
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Phase 3- Underlying Logic and Understanding Industry Standards and Principles

11. *"why havent we used industry standard formulas in this project?"*
12. *"mention all the formulas for water flow, flush count, pressure for all modes and the underlying hydraulic principles"*
13. *"while instantiating the model, how did we decide the number for decision trees, the percentage of anomalous data and what's random state exactly"*
14. *"give the formulas for calculating the kpis"*
15. *"How is isolation forest being used in this project?"*
16. *"The isolation forest is being retrained on every tick. Won't this cause data poisoning?"*
-

Phase 4- Code Generation using GPT

simulator.py - Synthetic Telemetry Generator

17. *"Task: Write a modular Python script simulator.py that generates continuous, synthetic IoT telemetry for a high-footfall commercial facility (e.g., Airport Terminal Restroom) to simulate Kohler commercial fixtures."*

Requirements:

1. Function `generate_telemetry_tick(zone_id, fixture_id, anomaly_mode=None)` that returns a JSON object.
2. Fields: `timestamp` (ISO format), `zone_id`, `fixture_id` (e.g., `KOHLER_FAUCET_101`), `water_flow_lpm` (float), `flush_count_10m` (int), `occupancy_count_10m` (int), `pressure_psi` (float), and `diagnostic_code` (str).
3. Modes:
 - **NORMAL:** Baseline water flow (0.0 - 1.5 LPM) strictly correlated with positive occupancy.
 - **LEAK:** Continuous flow (0.6 - 1.2 LPM) while occupancy is 0.
 - **TRAFFIC_SPIKE:** High occupancy (>100) and high flush counts over a short duration.

Include a test block under `if __name__ == '__main__':` that prints 5 ticks to stdout.”

analytics.py - Analytics & ML Engine

18. “Task: Write `analytics.py` in Python to process streaming telemetry JSON ticks and detect continuous micro-leaks and dynamic hygiene breaches.

Requirements:

Leak Detection Engine: Implement rolling baseline Z-score and Isolation Forest (`sklearn.ensemble.IsolationForest`) anomaly detection. Flag `ERR_CONTINUOUS_LEAK` if water flow stays >0.25 LPM for over 5 ticks while occupancy is 0.

Dynamic Hygiene Index: Calculate a score from 0 to 100 using:
 $\text{Hygiene Score} = 100 - (0.4 * \text{Occupancy}) - (0.5 * \text{Flush Count})$
 Flag `HYGIENE_BREACH` if the score drops below 30.

Provide clean helper functions `analyze_tick(tick_json)` returning an anomaly payload dict.”

app.py - Multi-Persona Dashboard

19. “Task: Write a complete `app.py` using Streamlit and Plotly to build the Kohler AquaSense OS dashboard.

Requirements:

Inject custom CSS at the top for a dark slate aesthetic (background #0b0f19, card headers #38bdf8, cards #1e293b).

Sidebar controls with buttons: [Inject Continuous Leak], [Simulate Flight Rush], [Reset System].

Tab 1: Executive View (Facility Head)

Top KPI cards: Water Saved (Liters), Cost Saved (\$), System Uptime (%).

Interactive Plotly 2D Spatial Floor Plan Map showing fixture positions (Sinks, Urinals, Stalls) colored Green (Normal), Amber (Hygiene Warning), or Red (Active Leak).

Live Anomaly Stream table.

Tab 2: Field Technician Portal (Marcus)

Active Work Order Cards displaying Ticket ID, Fixture Name, Kohler Part #, LLM Diagnostic Summary, and an [Acknowledge & Complete] button that clears the fault state.”

agent.py - LLM Work Order Agent System

20. “Task: Generate the production-grade System Prompt that will be hardcoded inside my agent.py module to call the Gemini API (gemini-2.5-flash or Groq API).

Requirements:

- Role: Kohler Commercial Facility Maintenance Intelligence Engine.
- Input: Anomaly payload JSON from analytics.py.
- Output: Enforce valid JSON ONLY matching this schema:

```
{
  "ticket_id": "str",
  "priority": "CRITICAL | HIGH | ROUTINE",
  "fixture_model": "str (e.g., Kohler Insight Touchless Faucet K-13491)",
  "root_cause_analysis": "str",
  "recommended_action": "str",
  "kohler_part_number": "str (e.g., Part #K-1138241)",
  "estimated_water_saved_lph": float
}
```

- **4. Guardrails:** No preamble, no postscript. The response must be programmatically parseable by `json.loads()`.

eval_agent.py

21. "i need an evaluation script that correctly evaluates my agent and gives the valid metrics, not anything merely made up. the output should be justifiable and accurate as per my project."

Phase 5- UI Redesigning- GPT

22. "im thinking to redesign the ui as streamlit looks too basic. is it feasible given the time constraint?"

main.py- FastAPI backend

23. "I am refactoring my Kohler AquaSense OS hackathon project from Streamlit to a Next.js (Frontend) + FastAPI (Backend) architecture.

I already have simulator.py, analytics.py, and agent.py working. Write a complete main.py script using FastAPI and uvicorn.

Requirements:

1. CORS Middleware enabled for local Next.js frontend (<http://localhost:3000>).
2. GET /api/telemetry: Returns full JSON state of 9 commercial fixtures (4 Sinks, 3 Urinals, 2 Stalls) with LPM flow, occupancy, status (NORMAL, LEAK, HYGIENE_WARNING), and active dynamic hygiene score.
3. POST /api/simulate: Accepts JSON payload `{ "action": "INJECT_LEAK" | "FLIGHT_RUSH" | "RESET" }` and updates global in-memory state.
4. POST /api/dispatch: Calls agent.py (Gemini LLM) for any active anomaly, returning structured JSON work orders matching this schema: ticket_id, fixture_id, priority, root_cause, recommended_action, kohler_part_number, water_saved_lph.
5. GET /api/kpis: Returns overall system stats (Total Water Saved Liters, Cost Saved USD, Active Leaks Count, System Uptime %).

Include all necessary imports and runnable uvicorn server block."

next.js + tailwind ui setup

24. *"I am building the Next.js 14 (App Router) frontend for Kohler AquaSense OS using Tailwind CSS and Lucide React icons.*

Write page.tsx and a Header/KPI component that matches a high-end dark slate commercial facility management system.

Requirements:

- 1. Dark Slate Theme (#0b0f19 background, #111827 card background, #1f2937 borders, #38bdf8 accents).*
- 2. Top Bar: Kohler Logo branding, "AquaSense OS - Commercial Facility Intelligence", and live status pill (System Online).*
- 3. Persona Switcher Tabs at top: [Executive View] and [Field Technician Portal - Marcus].*
- 4. Top KPI Metric Strip (3 cards):*
 - Water Saved (e.g., 1,248 L | +12.4% vs baseline)*
 - Cost Saved (e.g., \$18.72 | Est. daily operational savings)*
 - Active Fleet Uptime (e.g., 99.7% | 9 Connected Fixtures)*
- 5. Fetch data continuously from FastAPI endpoint `http://localhost:8000/api/kpis` using standard fetch or SWR.*

Give me clean, copy-paste-ready Next.js React code."

Interactive 2D Hydro-Spatial Floor Plan Map Component

25. *"Write a Next.js React component `SpatialFloorPlan.tsx` using Tailwind CSS that visualizes an interactive 2D Restroom Floor Map for Kohler commercial fixtures.*

Requirements:

- 1. Layout: A structured 2D grid layout simulating an Airport Terminal Restroom Complex:*
 - Top Zone: Sinks 1 to 4 (Hand-Wash Area)*
 - Middle Zone: Urinals 1 to 3*
 - Bottom Zone: Stalls 1 to 2*
- 2. Visual States for each fixture box:*
 - Green border/glow (#22c55e): Normal Operation*

- Amber border/glow (#f59e0b): High Traffic / Cleaning Needed
- Red border/glow (#ef4444) with pulse animation: Active Micro-Leak Detected

3. Interactivity: Clicking any fixture box opens a modal/drawer showing live telemetry details (Flow LPM, Occupancy, Pressure PSI, Last Cleaned).

4. Simulation Action Buttons bar at bottom:

- [Inject Solenoid Leak]
- [Simulate Flight Landing Rush]
- [Reset All Sensors]

These buttons trigger `POST http://localhost:8000/api/simulate`.

Provide full TypeScript React code.”

Field Technician Portal Component (Tab 2)

26. “Write a Next.js React component `TechnicianPortal.tsx` using Tailwind CSS for the Field Technician View (“Marcus”).

Requirements:

1. Card-based UI list fetching open work orders from `http://localhost:8000/api/dispatch`.

2. Card Details:

- Ticket ID Badge (e.g., #WO-30828) & Priority Badge (CRITICAL / HIGH / ROUTINE).
- Fixture Location & Name (e.g., Sink Bank A1 - KOHLER_FAUCET_101).
- Kohler Replacement Part Number highlighted in bold (e.g., Touchless Solenoid Kit #K-1138241).
- LLM-generated Root Cause Analysis & Recommended Repair Action.

3. Interactive Action: “Acknowledge & Complete” button on each card that sends a request to clear the ticket and resets that fixture’s state back to normal on the spatial map.

4. Empty state visual when zero tickets are active (“✓ All fixtures operating within normal parameters”).

Provide full TypeScript React code”

Phase 6- Running and Testing API Endpoints

27. *"sh: next: command not found
Even though npm is installed"*

28. *"give the steps to test the api endpoints"*

Phase - Presentation Deck- Claude

29. *"Draft the exact slide content for my Kohler AquaSense OS pitch deck (Max 4 Slides PDF).*

Technical context:

- Track 2: Smart Facility & Sustainability

- Stack: Next.js + Tailwind CSS (Frontend), FastAPI (Backend), Scikit-learn Isolation Forest (Anomaly ML), Gemini 2.5 Flash API (Agentic Dispatcher).

- Core Innovation: 2D Spatial Hydro-Map, continuous micro-leak detection, dynamic traffic hygiene index, and structured Kohler part-number dispatching.

Provide:

1. Slide 1 (Executive Overview & Problem Statement)

2. Slide 2 (System Architecture Diagram breakdown)

3. Slide 3 (AI/ML & Kohler Product Integration)

4. Slide 4 (Business Impact & Water Stewardship Metrics)

Phase 9- Changing data according to PNQ Airport

30. *"tell me what all i need to change across all the files including all the data to make it according to pune international airport"*

Phase 10- Final Testing

31. *"2 tickets are being injected instead of 1" [Attached screenshot]*

32. *"ML anomaly tickets rising up even though no leak or traffic spike was injected"*
